

Thomas Andrew Lamb

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Curriculum Vitæ

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Research Overview: My general research interests concern AI safety and reliable AI systems. In particular, how can we elicit, produce and give guarantees for the safe and reliable deployment of ML models, specifically LLMs and variants such as reasoning models, in high-stakes environments. This work has led to several peer-reviewed works on topics such as uncertainty quantification for LLMs, steering LLMs to improve robustness and reduce biases at inference-time, biases of VLMs, and universal-approximation theory for LLMs. Currently I am particularly interested in uncertainty quantification, specifically calibration, of LLMs, as well as Bayesian and approximate inference methods.

EDUCATION

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- University of Oxford** Oxford, UK
PhD (DPhil) in Machine Learning, Department of Engineering Science Oct. 2023 – Present
 - Supervised by Dr. Tim G. J. Rudner (University of Toronto & Vector Institute) and Prof. Philip H.S. Torr (University of Oxford).
 - Published at top-tier ML conferences including ICML, NeurIPS, AISTATS and ICLR. These include papers on: the semantic calibration of and uncertainty quantification for LLMs; steering LLMs adaptively at inference time; biases in VLMs; and giving formal statements on universal in-context approximation of fully recurrent models including SSMs.
 - University of Edinburgh** Edinburgh, UK
MSc Artificial Intelligence (Distinction) Sept. 2022 – Aug. 2023
 - Courses taken included: Reinforcement Learning, Bayesian Statistics and MCMC Methods, Probabilistic Modeling and Reasoning, and Natural Language Understanding, Generation and Machine Translation. Audited Targeted Causal Learning, Methods of Causal Inference, and Algorithmic Game Theory.
 - MSc dissertation: ‘Self-Supervised Learning of Tractable Generative Models’, supervised by Dr. Antonio Vergari. Applied conditional composite log-likelihood estimation (CCLE) with novel patching schemes to train EiNet models. Demonstrated that training using CCLE with specific patching schemes shows promise for improved image inpainting performance.
 - University of Durham** Durham, UK
MMath Mathematics (First-Class Honours) Oct. 2016 – July 2020
 - Specialised in pure mathematics. Selection of courses taken: Algebraic Topology, Riemannian Geometry, Representation Theory, Partial Differential Equations, Algebraic Number Theory, Galois Theory, Statistics, Analysis in Many Variables, Algebra, and Numerical Analysis.
 - MMath dissertation: ‘An Introduction to Modular Forms and the Eichler–Shimura Isomorphism’, supervised by Prof. Herbert Gangl. Provided a comprehensive introduction to the theory of classical Modular Forms. This included detailed discussions on Hecke operators, Maeda’s conjecture, L -functions and was concluded by proving the Eichler–Shimura Isomorphism.

SELECTED PUBLICATIONS AND PREPRINTS

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- Lamb, T.A., Ivanova, D.R., Torr, P. and Rudner, T.G., 2025. Improving Semantic Uncertainty Quantification in Language Models via Token-Level Temperature Scaling. AISTATS, 2025.
 - Stoisser, J.L., Phillips, L., Misra, A., Lamb, T.A., Torr, P., Martell, M.B., Fauqueur, J. and Mörtens, K., 2025. Towards Label-Free Biological Reasoning Synthetic Dataset Creation via Uncertainty Filtering. NeurIPS 2025 ER Workshop.
 - Tao, L., Yeh, Y.F., Kai, B., Dong, M., Huang, T., Lamb, T.A., Yu, J., Torr, P.H. and Xu, C., 2025. Can Large Language Models Express Uncertainty Like Human?. arXiv preprint arXiv:2509.24202.
 - Lamb, T.A., Ivanova, D.R., Torr, P. and Rudner, T.G., 2025. Semantic-Level Confidence Calibration of Language Models via Temperature Scaling. In ICLR Workshop: Quantify Uncertainty and Hallucination in Foundation Models: The Next Frontier in Reliable AI, 2025.
 - Kim, H., Lamb, T.A., Bibi, A., Torr, P. and Gal, Y., 2024. Detecting llm hallucination through layer-wise information deficiency: Analysis of unanswerable questions and ambiguous prompts. ENLP, 2025
 - Lamb, T.A., Davies, A., Paren, A., Torr, P. and Pinto, F., Focus On This, Not That! Steering LLMs with Adaptive Feature Specification. In Forty-second International Conference on Machine Learning, 2025.
 - Hemmat, A., Davies, A.*, Lamb, T.*, Yuan, J.*, Torr, P., Khakzar, A. and Pinto, F., 2024. Hidden in plain sight: evaluating abstract shape recognition in vision-language models. Advances in Neural Information Processing Systems, 37, 2024.
 - Petrov, A., Lamb, T., Paren, A., Torr, P. and Bibi, A., 2024. Universal in-context approximation by prompting fully recurrent models. Advances in Neural Information Processing Systems, 37, pp.72061-72093.
 - Lamb, T. A., Brunel, R., Dvijotham, K. D., Kumar, M. P., Torr, P. H., & Eiras, F. (2023). “Faithful Knowledge Distillation”. arXiv preprint arXiv:2306.04431.

PRIZES

- **MSc Artificial Intelligence Class Prize (School of Informatics, University of Edinburgh):** Awarded to the student who attained the highest overall mark in the MSc Artificial Intelligence course.
- **The Percy Heywood Prize (Department of Mathematical Sciences, University of Durham):** Awarded to a student graduating with an MMath Master of Mathematics whose performance is outstanding in the final year.
- **The John Crowther Prize (University College, University of Durham):** Awarded by University College for performance in my third year Mathematics exams.

RESEARCH EXPERIENCE AND INTERNSHIPS

- **Spring into Quant Finance 2025** Saint-Jean-Cap-Ferrat, France
G-Research April 2025
 - Selected via competitive entrance to G-Research's week-long Quant Finance program, with workshops on mathematical finance, ML in finance, and hands-on coding hackathons.
- **ELLIS Summer School on Probabilistic Machine Learning** Cambridge, UK
University of Cambridge July 2024
 - Participated in workshops on ML model calibration and uncertainty quantification, Bayesian methods in ML, causal machine learning, AI for science, and geometric deep learning.
- **DeepMind Machine Learning Research Intern** Oxford, UK
University of Oxford (Torr Vision Group) July 2022 – January 2023
 - Research intern funded by Google DeepMind and supervised by Francisco Girbal Eiras and Prof. Philip H. S. Torr.
 - Introduced a faithful-imitation framework for teacher–student networks, formulated LP-based upper/lower bounds on measures of the local confidence disagreement between teacher and student models, and designed a distillation method yielding verifiably more calibrated student models.
 - Lead to our paper “Faithful Knowledge Distillation”.
- **University of Durham** Durham, UK
Summer Research Student July 2019 – Sept. 2019
 - Collaborated with Prof. Herbert Gangl to conjecture and prove new identities relating polylogarithmic integrals over the unit square to linear combinations of multiple zeta values.

TALKS

- **Improving Semantic Uncertainty Quantification in LLMs via Token-Level Temperature Scaling** Sept. 2025
Novo Nordsik. Oxford, UK
- **Semantic-Level Confidence Calibration of Language Models via Temperature Scaling** Mar. 2025
LLM Seminar, Dept. of Statistics, University of Oxford. Oxford, UK
- **Towards Safety and Transparency: Addressing Bias and Uncertainty Quantification in LLMs** Oct. 2024
Huawei Noah's Ark Lab. London, UK

TEACHING

- **OXAi Education Team Member** Oxford, UK
University of Oxford Sep. 2023 – June 2024
 - Created blog posts and teaching materials on machine learning, ranging from introductory concepts to advanced topics for both undergraduates and researchers.
- **Mathematics and AI Tutor** Oxford, UK & Remote
University of Oxford and Online Sep. 2020 – Jan. 2025
 - Teach MSc-level Reinforcement Learning (OPUS program) and tutor GCSE/A-Level and undergraduate mathematics.
- **Mathematics Teaching Assistant** Durham, UK
University of Durham Oct. 2019 – June 2020
 - Marked and provided detailed feedback on second-year Algebra coursework, ensuring clarity and rigor for student learning.

TECHNICAL SKILLS

- **Machine Learning:** NLP, LLMs, representation learning, variational methods, deep learning, Bayesian methods.
- **Programming/Frameworks:** Python, Bash, Git, WandB, Java, Matlab, SQL.
- **Specific Python Packages Used:** PyTorch, GPyTorch, Transformers, OpenAI, Pandas, GurobiPy, SciPy, Scikit-learn, NLTK, Datasets, NumPy.